

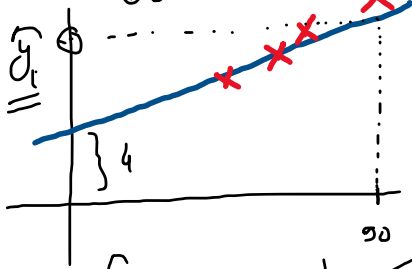
Perceptron Trick

02 November 2025 12:25

Linear regression predicts a continuous outcome, while logistic regression predicts the probability of a categorical outcome. The main difference is their application: linear regression is for "how much" or "how many" questions (e.g., predicting house price), and logistic regression is for "yes/no" or "which category" questions (e.g., predicting if an email is spam)

Linear Reg ← Regression

marks	placement
80	8L
77	7L
62	5L
90	10L
68	72



$$\hat{y}_i = mx_i + b$$

Value - ? $\hat{y}_i - mx_i - b = 0$

$$\hat{y}_{68} = m \cdot 68 + b$$

$$= 1.68 + 4$$

$$= 72$$

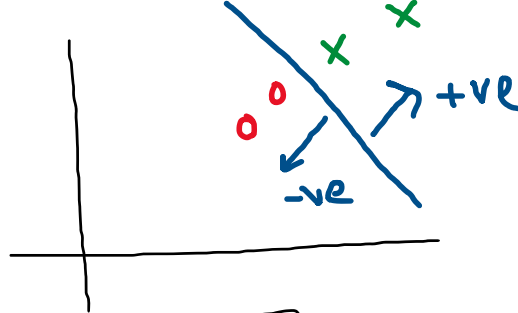
$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

$$\hat{y} = \sum_{i=0}^n w_i x_i \mid x_0 = 1$$

Raff

Logistic Reg

marks	placement
80	8L 1
77	7L 0
62	5L 0
90	10L 1



$$Ax + By + C = 0$$

$$(Ax + By + C)(x_1, y_1) > 0 \quad +ve$$

$$< 0 \quad -ve$$

$$= 0 \quad \text{on the line}$$

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

$$\hat{y} = \sum_{i=0}^n w_i x_i \mid x_0 = 1$$

$$\sum_{i=0}^n w_i x_i \leftarrow \text{value}$$

$$= 0$$

$$> 0$$

$$y = f(x_1, x_2, \dots, x_n)$$

classification

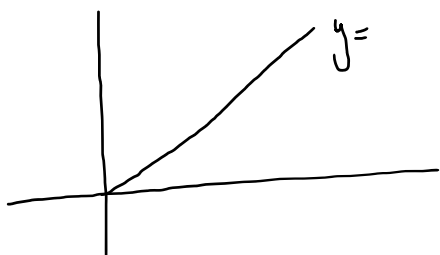
categorical outcome
class → yes/no

+ve/-ve

Raff

$$y = mx + b$$

$$y - mx - b = 0 \leftarrow \text{line eq}^n$$



$$Ax + By + C = 0 \quad || \quad y = -\frac{A}{B}x - \frac{C}{B}$$

$$w_0 + w_1x_1 + w_2x_2 = 0$$

2D $\nearrow$

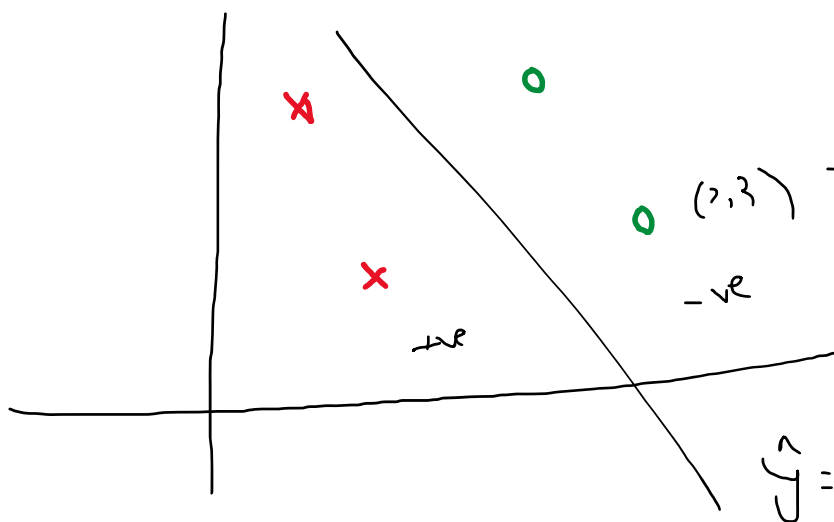
$$3D \quad w_0 + w_1x_1 + w_2x_2 + w_3x_3 = 0$$

$$\boxed{y = mx - b}$$

$$m = -\frac{A}{B}$$

$$b = -\frac{C}{B}$$

$$nD \quad w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n = 0$$



$$(2, 3) \quad -4 \cdot 2 - 3 - 5 = -8 - 11 = -19$$

-ve

$$\boxed{-4x - \hat{y} - 5}$$

$$\hat{y} = 4x + 5$$

$$\Rightarrow \boxed{-4x - \hat{y} - 5 = 0}$$

- i) +ve p $\rightarrow$ -ve reg(+)
- ii) -ve p $\rightarrow$ +ve reg(-)

for i in epochs:-

randomly select x_i :-

if) $x_i \in +ve$ and

$$\sum_{i=0}^n w_i x_i = -ve \frac{\partial}{\partial r}$$

$$w = w + \eta x_i$$

it) $x_i \in -ve$ and $\sum_{i=0}^n w_i x_i = +ve$:-

$$w = w - \eta x_i$$

for i in epochs:-

randomly selecting x_i

$$w = w + \eta (y_i - \hat{y}_i) x_i$$

substitute

y	$\hat{y}$	$y - \hat{y}$
0	0	0
0	1	-1
1	1	0
1	0	1

Disadvantage:-

i) randomness

ii) Partiality

11) Positivity

$$\sum_{i=1}^n w_i x_i$$

$$= \underbrace{w_0 + \sum_{i=1}^n w_i x_i}_{> 0}$$

Sigmoid Function

02 November 2025

14:25

Linear Regression

↓
OLS

↓
GP (3) ✓

$$\hat{y} = \sum_{i=0}^n w_i x_i \quad \text{col\#}$$

Loss function (1 row)
Cost function (Dataset)

$$\frac{1}{n} \sum_{i=0}^n (y_i - \hat{y}_i)^2$$

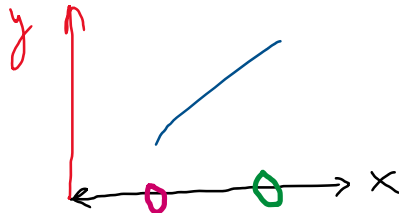
$$W_n = W_0 - \eta \frac{dL}{dW_0}$$

i) $y_i - \hat{y}_i \approx 0$ ✓

ii) loop : 1: epoch ✓

$$\begin{matrix} 0.01 \\ 0.0001 \end{matrix} \quad \times$$

Logistic Regression:-



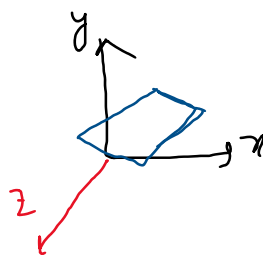
Classification

$$C = y$$

$$y = f(x) = x^2$$

$$y = x^2$$

$$z = f(x, y) = x^2 + y^2$$

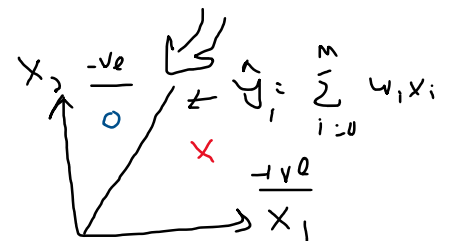


i/p			
x_1	x_2	y	$\hat{y}_i$

Perceptron trick

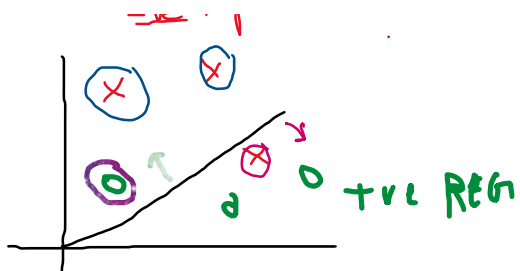
for i in range (epoch):-

$$w_m = w_0 + \eta (y_i - \hat{y}_i) x_i$$



$$\hat{y}_i = \text{step} \left(\sum_{i=0}^n w_i x_i \right)$$

✓ Reg
| (x) (x)



$$x_i \in N \text{ of } \sum_{i=0}^n w_i x_i \geq 0$$

$$w_n = w_0 - \eta x_i$$

$$x_i \in P \text{ of } \sum_{i=0}^n w_i x_i < 0$$

$$w_n = w_0 + \eta x_i$$

y_i	$\hat{y}_i$
0	0
0	1
1	0
1	1

$$w_n = w_0$$

$$w_n = w_0 - \eta x_i$$

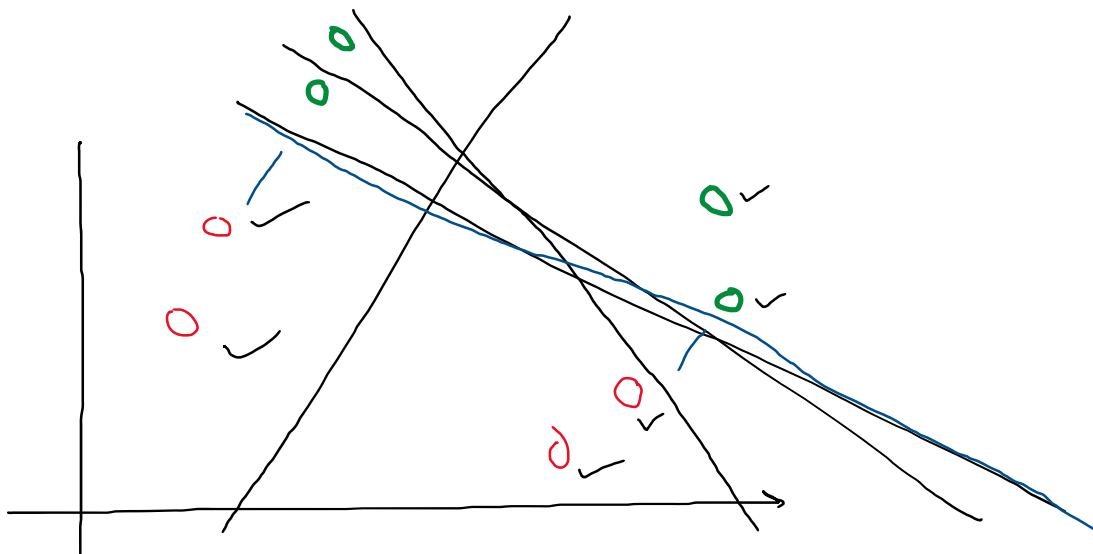
$$w_n = w_0 + \eta x_i$$

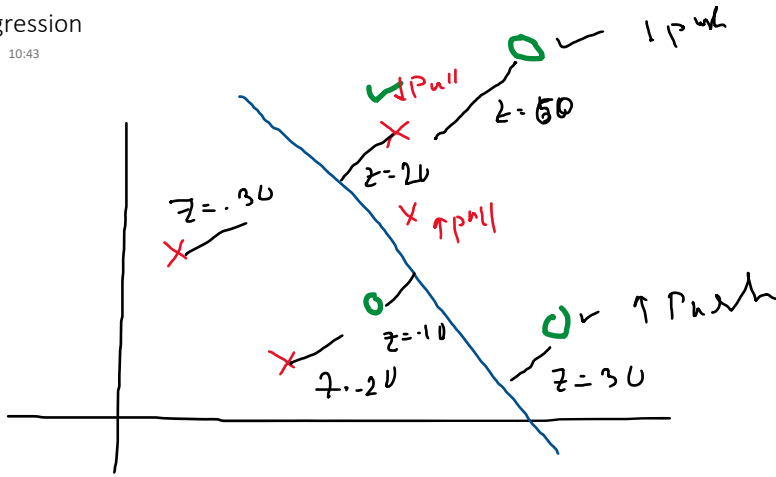
$$z = f(a)$$

$$0 \leq a < 0$$

$$1 \leq a > 1$$

Problem





$$z = \sum_{i=0}^n w_i x_i$$

$$z = \sum_{i=0}^n w_i x_i$$

P.T

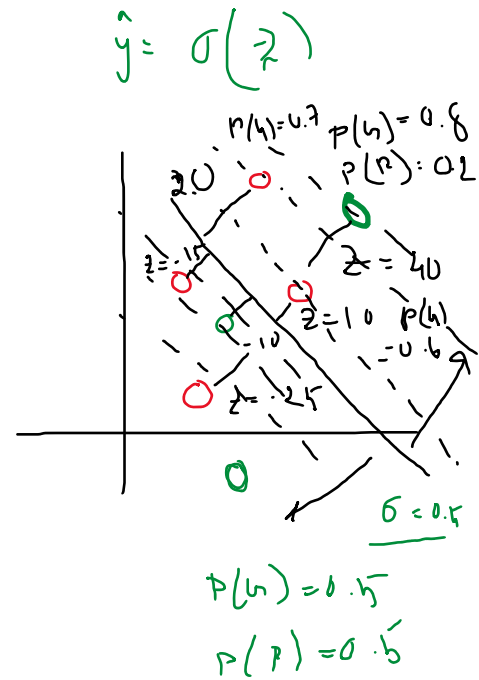
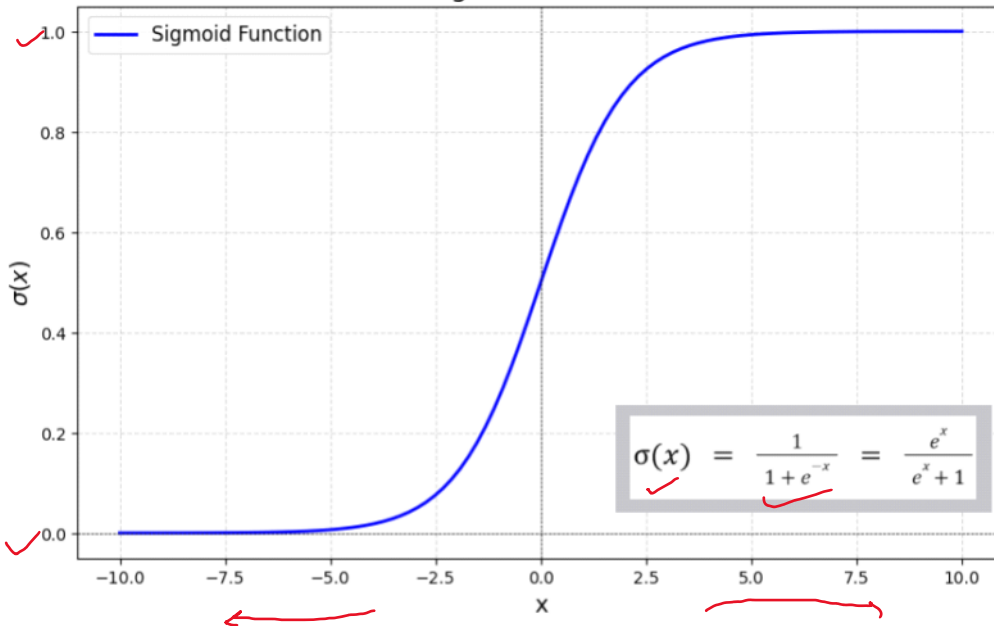
$$\hat{y}_i = \text{step}(z)$$

Probability: =

$$z = \sum_{i=0}^n w_i x_i$$

$$\hat{y}_i = \sigma(z)$$

Sigmoid Function



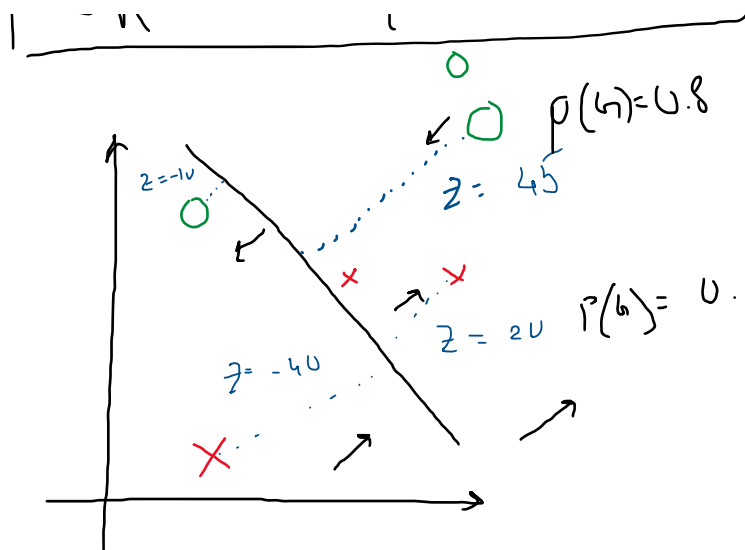
$$\sigma(z) \geq 0.5 \quad \textcircled{1} \text{ Green}$$

$$\sigma(z) < 0.5 \quad \textcircled{0} \text{ Red}$$

$$w_n = w_0 + \eta (y_i - \hat{y}_i) x_i$$

$\hat{y}_i = \sigma(w_n x_i)$

$$y_i \quad | \quad \hat{y}_i \quad | \quad y_i - \hat{y}_i$$



y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	0.8	0.2
0	0.6	-0.6
1	0.4	0.6
0	0.2	-0.2

$$i) \quad w_n = w_0 + \eta \cdot 0.2 \cdot x_i$$

$$ii) \quad w_n = w_0 - \eta \cdot 0.6 \cdot x_i$$

$$iii) \quad w_n = w_0 + \eta \cdot 0.6 \cdot x_i$$

$$iv) \quad w_n = w_0 - \eta \cdot 0.2 \cdot x_i$$

(magnitude $\propto$ distance)

$$z_1 = w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n}$$

$$z_2 = w_0 + w_1 x_{21} + \dots + w_n x_{2n}$$

$$z_m = w_0 + w_1 x_{m1} + \dots + w_n x_{mn}$$

$$Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_m \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}$$

$$= X \cdot W = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} + w_0$$

$$x \rightarrow 100, 23$$

$$w = [1, 3]$$

$$x[j] \rightarrow [1, 3]$$

$$(1, 3) \quad (? , 1)$$

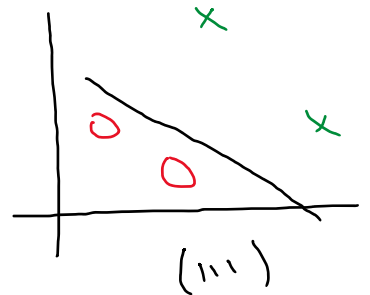
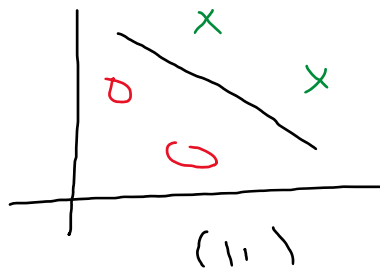
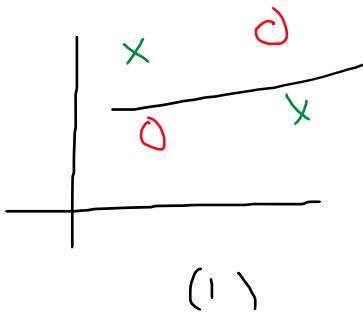
$$= 1$$

$$z_j = w_0 + w_1 x_{j1} + w_2 x_{j2}$$

$$= \begin{bmatrix} 1 & x_{j1} & x_{j2} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

Problem:-

Randomness



Machine learning:-

OLS

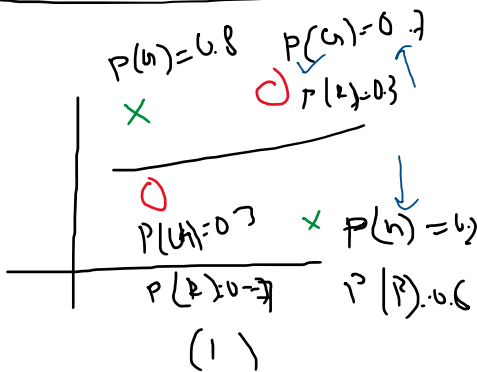
$$\frac{dL}{dW} = 0$$

GD

$$W_n = W_0 - \eta \frac{dL}{dW}$$

Cost function $\leftarrow$ Minimize

L minimum cost $\leftarrow$ minimize

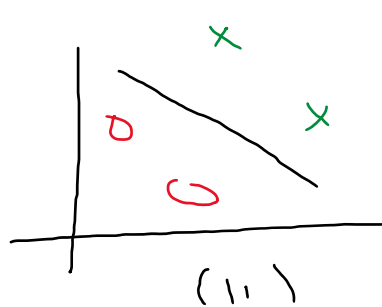
$$C = f(\underbrace{w_0}_{\sim}, \underbrace{w_1}_{\sim}, \underbrace{w_n}_{\sim})$$


$$0.8 \times 0.3 \times 0.7 \times 0.2$$

$$= 0.0336$$

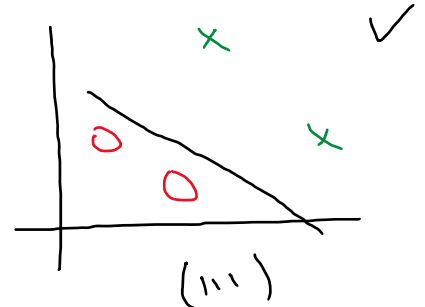
maximum likelihood

Problem



$$= 0.2016$$

(maximum)



$$= 0.3024$$

$$= 0.3024$$

$$8 \times 8 = 64$$

$$8 \times 0.5 = 4$$

1 Problem

$$\frac{100}{8 \times 0.5} = 4$$

$$\frac{0.5 \times 0.5}{0.5} = 0.25$$

$$\log(ab) = \log a + \log b$$

m1

m2

$$\log(0.7) + \log(0.6) + \log(0.6) + \log(0.7)$$

m3

$$-\log(0.7) - \log(0.6) - \log(0.6) - \log(0.7)$$

cross on top y

(minimum)

$$\log(0.1) = -1$$

$$= -\log(0.1) > -\log(0.9) > 0.04$$

$$\log(0.9) = 0.04$$

$$= 1$$

$$\begin{matrix} y & \hat{y} \\ y_1 & \hat{y}_1 \\ y_2 & \hat{y}_2 \\ \vdots & \vdots \end{matrix}$$

$$-\log \hat{y}_1 - \log \hat{y}_2 - \log \hat{y}_3 - \log \hat{y}_n \quad X$$

81

$$1 \cdot \hat{y} \cdot X$$

$$\begin{array}{cc}
 y_3 & \hat{y}_3 \\
 y_n & \hat{y}_n \\
 \hline
 y & \hat{y} \\
 \hline
 1 & 0.7 \\
 0 & 0.3 \\
 \hline
 \end{array}$$

$$1^{st} \rightarrow -\log \hat{y}_1 \times$$

$$\begin{aligned}
 & -y_1 \log(\hat{y}_1) - \underline{(1-y_1) \log(1-\hat{y}_1)} \\
 & = -1 \log(0.7) - \underline{(1-1) \log(1-0.7)} \\
 & = -\log(0.7) \\
 & = -0 \log(0.3) - (1-0) \log(1-0.3) \\
 & = -\log(0.3)
 \end{aligned}$$

$$L_{\text{avg}} = -y_i \log \hat{y}_i - (1-y_i) \log(1-\hat{y}_i)$$

$$\text{Cost function} = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y}_i) \right]$$

OLS

GD

$$w = w - \eta \frac{dL}{dw}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\frac{d}{dx} x^{-1} = -\frac{1}{x^2}$$

$$\sigma'(x) = \frac{d}{dx} \left[\frac{1}{1 + e^{-x}} \right] = \frac{d}{dx} \left((1 + e^{-x})^{-1} \right) \left| \begin{array}{l} \frac{d}{dx} y^2 \\ = 2y \frac{dy}{dx} \end{array} \right.$$

$$= -\frac{1}{(1 + e^{-x})^2} \cdot \frac{d}{dx} (1 + e^{-x})$$

$$= -\frac{1}{(1 + e^{-x})^2} \cdot e^{-x} \cdot \frac{d}{dx} (-x)$$

$$\frac{d}{dx} e^x = e^x$$

$$= -\frac{1}{(1 + e^{-x})^2} \cdot e^{-x} \cdot (-1)$$

$$= \frac{e^{-x}}{(1 + e^{-x})^2}$$

$$= \left(\frac{1}{1 + e^{-x}} \right) \frac{e^{-x}}{(1 + e^{-x})} = \sigma(x) \frac{1 + e^{-x} - 1}{1 + e^{-x}}$$

$$= \sigma(x) \left[\frac{1 + e^{-x}}{1 + e^{-x}} - \left(\frac{1}{1 + e^{-x}} \right) \right]$$

$$\sigma'(x) = \sigma(x) [1 - \sigma(x)]$$

$$C = -\frac{1}{m} \sum_{i=1}^m (y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y}_i))$$

$$\hat{y}_1 = \sigma(w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n} + w_0)$$

$$\hat{y}_2 = \sigma(w_1 x_{21} + \dots + w_n x_{2n} + w_0)$$

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} \sigma(w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n} + w_0) \\ \sigma(w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n} + w_0) \\ \vdots \\ \sigma(w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn} + w_0) \end{bmatrix}$$

$$= \sigma \left(\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \right)$$

$$= \sigma(XW)$$

$$\boxed{\hat{Y} = \sigma(XW)}$$

$$C = -\frac{1}{m} \sum_{i=1}^m (y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y}_i))$$

$$C = -\frac{1}{n} \sum_{i=1}^n \log y_i$$

$$= -\frac{1}{n} \left(\sum_{i=1}^n y_i \log \hat{y}_i + \sum_{i=1}^n \log(1 - \hat{y}_i) \right)$$

$$\sum_{i=1}^n y_i \log \hat{y}_i = y_1 \log \hat{y}_1 + y_2 \log \hat{y}_2 + \dots + y_n \log \hat{y}_n$$

$$= [y_1 \ y_2 \ \dots \ y_n] \begin{bmatrix} \log \hat{y}_1 \\ \log \hat{y}_2 \\ \vdots \\ \log \hat{y}_n \end{bmatrix}$$

$$= [y_1 \ y_2 \ \dots \ y_n] \log \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$$= Y \log \hat{Y}$$

$$= Y \log(\sigma(xw))$$

$$\sum_{i=1}^n (1 - y_i) \log(1 - \hat{y}_i)$$

$$= [(1 - y_1) \ (1 - y_2) \ \dots \ (1 - y_n)] \log \begin{bmatrix} 1 - \hat{y}_1 \\ 1 - \hat{y}_2 \\ \vdots \\ 1 - \hat{y}_n \end{bmatrix}$$

$$= (1 - Y) \log(1 - \hat{Y})$$

$$= (1-y) \log(1-y)$$

$$= (1-y) \log(1 - \sigma(xw))$$

$$\Rightarrow C = -\frac{1}{n} \left[Y \log \hat{Y} + (1-Y) \log(1-\hat{Y}) \right]$$

where $\hat{Y} = \sigma(xw)$

$$w = [w_1 \quad w_2 \quad w_3 \quad \dots \quad w_n]$$

$$\underline{w} = \underline{w} - \eta \underline{\frac{dL}{dw}} \quad \left| \quad \frac{dL}{dw} = \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right] \right.$$

$$\frac{dL}{dw} = \frac{d}{dw} \left[-\frac{1}{n} \left(Y \log \hat{Y} + (1-Y) \log(1-\hat{Y}) \right) \right]$$

$$\frac{d}{dw} (Y \log \hat{Y}) = \frac{d}{dw} (Y \log \sigma(xw)) \quad \left| \quad \frac{d}{dz} \log z = \frac{1}{z} \right.$$

$$= Y \frac{d}{dw} (\log(\sigma(xw)))$$

$$= \frac{Y}{\sigma(xw)} \frac{d}{dw} \sigma(xw) \quad \left| \quad \frac{d}{dz} \sigma(z) = \sigma(z)(1-\sigma(z)) \right.$$

$$\begin{aligned}
 & \frac{d}{dw} \left[\frac{Y}{\cancel{\hat{Y}}} \underbrace{\sigma(xw)}_{\sigma(xw)} [1 - \underbrace{\sigma(xw)}_{\sigma(xw)}] \right] \cdot \frac{d}{dw} (xw) \\
 & = Y(1 - \hat{Y}) X
 \end{aligned}$$

$$\begin{aligned}
 & \frac{d}{dw} (1 - Y) \log(1 - \hat{Y}) \\
 & = \frac{d}{dw} (1 - Y) \log(1 - \sigma(xw))
 \end{aligned}$$

$$= -\hat{Y}(1 - Y) X$$

$$\frac{dL}{dw} = -\frac{1}{n} \left[Y(1 - \hat{Y}) X - \hat{Y}(1 - Y) X \right]$$

$$= -\frac{1}{n} \left[Y(1 - \hat{Y}) - \hat{Y}(1 - Y) \right] X$$

$$= -\frac{1}{n} \left[Y - \underline{Y\hat{Y}} - \hat{Y} + \underline{Y\hat{Y}} \right] X$$

L

$$\frac{dL}{dw} = -\frac{1}{n} (y - \hat{y}) x$$

$$w = w - \eta \left(-\frac{1}{n} (y - \hat{y}) x \right)$$

$$w = w + \eta \cdot \frac{1}{n} \cdot (y - \hat{y}) x$$